

PRANITA AGRAWAL

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Software Development Engineer with in-depth expertise in C++, Java, and Python programming languages, proficient at developing Rest APIs and working with MySQL and PostgreSQL databases. 3+ years of experience in the Banking domain with a strong understanding of Python programming, AWS, SQL, Flask libraries.

EXPERIENCE

Software Developer Intern (Leidos | USA)

June 2024 – August 2024

- Developed a Java microservices application on AWS for the Employer Income Withholding Order system, achieving a 50% reduction in server costs through efficient resource allocation and streamlined backend processes.

Student Assistant (Wayne State University | USA)

April 2023 - May 2024

- Spearheaded the creation of an advanced urban studies survey platform using ASP.NET and C++, implementing optimized stored procedures to automate data retrieval, resulting in a 40% increase in data processing efficiency.
- Configured AWS cloud services, including EC2 for scalable computing, S3 for secure storage, and RDS for managed database solutions.

Software Engineer (Tata Consultancy Services | India)

August 2021 - December 2022

- Enhanced ETL process efficiency by 30%, by implementing automated file processing using Python scripts and reduced overall processing time by 80%, using Informatica PowerCenter to develop ETL workflows.
- Managed SQL databases by developing precise SQL queries and stored procedures to ensure data accuracy by 99%.
- Built and automated data processing Python scripts for monthly files, ensuring seamless data loading with less than 0.1% downtime. Excelled the SLA time by providing continuous support in case of job/process failures.

Software Engineer Intern (Tata Consultancy Services | India)

February 2021 - July 2021

- Designed and deployed a straightforward Linear Regression model using the PySpark module, enabling students to predict likelihood of graduate admission with an accuracy of 92%.
- Streamlined the data validation process by implementing advanced data cleaning techniques, resulting in the identification of 3,000 duplicate records, allowing for a more reliable dataset for ongoing analysis and reporting.

EDUCATION

Master in Computer Science (Wayne State University | Detroit, MI | GPA: 3.91/4.0)

January 2023 - May 2024

Web Technology, C++, Cyber Security, Data Science, Microservices, Computer Systems Security, Machine Learning

Bachelor in Computer Science and Engineering (GVPCOE | India | GPA: 3.98/4.0)

August 2017 - July 2021

Data Structures and Algorithms, Object-Oriented Programming, Artificial Intelligence, DBMS, Software Engineering.

TECHNICAL SKILLS

Programming Languages: C++, Java, JavaScript, Python, SQL, HTML, CSS, ASP.Net

Web Technologies and Databases: MySQL, MongoDB, PostgreSQL, Docker, Kubernetes

Frameworks and Libraries: Flask, Pandas, PySpark, Docker, Spring, Rest API, AWS

PROJECTS

Tree-Cutting Contractor Management System (Wayne State University | December 2023)

MySQL | Java

- Engineered a Java application that enabled role-based access for 30+ users, incorporating functionalities for quote submission and order processing; improved user experience led to a 25% increase in user satisfaction ratings.

Inventory Management System (Wayne State University | April 2023)

Java | PostgreSQL

- Implemented a Java-based desktop application for inventory management, utilizing a microservices architecture.
- Integrated five APIs for seamless stock management, with PostgreSQL ensuring efficient record-keeping. Employed multi-threading and efficient memory management techniques to optimize application performance.

AI-Enhanced Cybersecurity System (Wayne State University | March 2023)

Cyber Security | Python

- Developed an AI-powered cybersecurity system in Python using Random Forest models for advanced threat detection. Optimized anomaly detection algorithms and integrated the system into existing frameworks, improving defense and response against evolving cyber threats.

Sentimental Tweets Classifier (GVPCOE | July 2021)

NLTK | API | Python

- Designed a web application to classify Twitter tweets by performing sentimental analysis using supervised learning Classification algorithm Naive Bayes and web framework Flask with an accuracy of 90%.
- Extracted 10k sentimental tweet data containing words like kill, die, suicide from Twitter using TWINT API.

CERTIFICATIONS

- Cyber Security Certification: ISSO: Certified Information Systems Security Officer.
- Coursera: Introduction to Data Science in Python.

AWARDS

- Secured 1st position in a coding contest conducted by TCS (All India).
- Awarded the "Infosys Certified Software Programmer" title in the Infyqt coding challenge by Infosys.